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Date	November 21, 2000

Ellipse Parameters Conversion and Vertical Velocity Profiles for Tidal Currents

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November 21, 2000

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Conversion between the tidal current amplitude and phase lag parameters (for short, referred to as ap-parameters hereafter) and tidal current ellipse parameters (referred to as e-parameters hereafter) is not as trivial as conversion between Cartesian and polar coordinates. We spend time to figure it how to do so at one time and then forget it in a few months later (or shorter, my e-folding memory scale is short, how long is yours?). I have just completed a tidal data assimilation project, in which I have done tons of such conversions with a sketchy MATLAB program. Recently some my colleagues inquired on how to do the conversion. Given that such inquiries are heard from time to time, I decided to pull all the relevant material together in one place for convenience in our future work. The rest of this document consists of two parts: a theory on the conversion and several MATLAB programs (version 5 or higher).

As you will see, having gone through the rotary decomposition for tidal ellipse parameters, it would be a waste if I did not go a step further for decoupling the linear tidal momentum equations, for the setting for the two cases are the same, and any one who is interested in tidal ellipse parameters is likely also interested in the tidal momentum equations. Having presented the decoupled momentum equations, I might go another mile to present a solution for the equations. Thus, the main body of the document consists 1) theories for the ellipse conversions and decoupling momentum equations, and 2) programs for conversion between ap- and e- parameters (ap2ep.m and ep2ap.m) and for calculation of vertical tidal current profiles (cBEpm.m) and the associated auxiliary programs. An example program (example.m) is also included to demonstrate how to use ap2ep.m and ep2ap.m.

This document and associated programs are a revision of my first release made a few weeks ago (you may regard that release as a beta-version if you already have down-loaded it). After that release, I received good response from Dr. Rich Signell of U.S. Geological Survey, who not only debugged the program but also gave me many good suggestions, especially on the designs of the notation for the minus rotary components. I enjoyed discussions with him and would like to express my sincere thanks to him. I also thank my colleague in BIO, Dr. Charles Hannah, for having a quick proofread of this document.

One more bit before I get into the real business, in putting up the mathematics, I chose a way that will make readers feel effortless in reading most of the derivations while still manage to keep the document from being fat. This contrasts to some traditional treatments for expressing mathematics, where readers are expected to read with pencils and scratch paper. If there is any one out there being

offended by seeing too much details, please forgive me!

1 Theory

1.1 Tidal ellipse and rotary components

Given tidal currents of u - (east or x-) and v - (north or y-) components, as

$$u = a_u \cos(\omega t - \phi_u) \quad (1)$$

$$v = a_v \cos(\omega t - \phi_v) \quad (2)$$

where a_u and ϕ_u are the amplitudes and phase lags for the u -components and likewise for a_v and ϕ_v , and ω is the tidal angular frequency, we can form a complex tidal current w as

$$w = u + iv \quad (3)$$

where $i = \sqrt{-1}$. If we trace w on a complex plane as time goes by a period ($T=2\pi/\omega$), we will see an ellipse. Our interest here is not only in seeing the ellipse, but also would like have the following ellipse parameters calculated:

- Maximum current velocity or semi-major axis (referred to as SEMA hereafter where appropriate).
- Eccentricity (ECC), the ratio of semi-minor to semi major axis, negative values indicating that the ellipse is traversed in a clockwise rotation;
- Inclination (INC), or angle between east (x-) and semi-major axis;
- Phase angle (PHA), i.e., the time of maximum velocity with respect to a chosen origin of time (if the phase lag is relative to Greenwich time, then the time will be also the Greenwich time).

Let us continue from eq. 3:

$$\begin{aligned} w &= u + iv \\ &= a_u \cos(\omega t - \phi_u) + ia_v \cos(\omega t - \phi_v) \\ &= a_u \frac{e^{i(\omega t - \phi_u)} + e^{-i(\omega t - \phi_u)}}{2} + ia_v \frac{e^{i(\omega t - \phi_v)} + e^{-i(\omega t - \phi_v)}}{2} \\ &= \frac{a_u e^{-i\phi_u} + ia_v e^{-i\phi_v}}{2} e^{i\omega t} + \frac{a_u e^{i\phi_u} + ia_v e^{i\phi_v}}{2} e^{-i\omega t} \end{aligned} \quad (4)$$

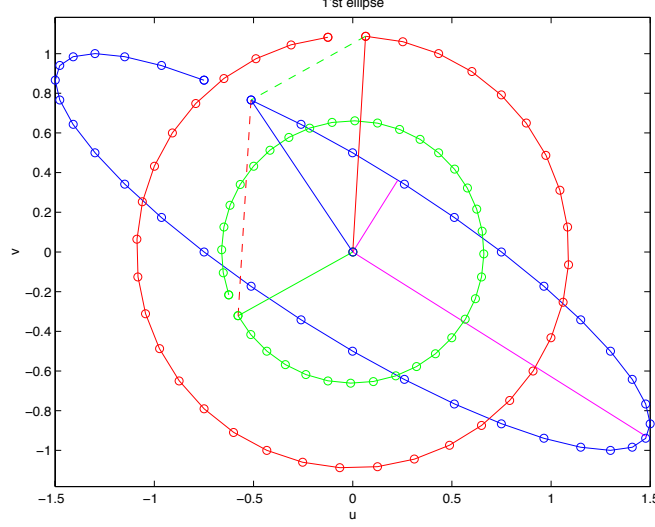


Figure 1: An ellipse can be constructed by two opposite rotating circles (red: anti-clockwise circle, green: clockwise circle, blue: ellipse). The circle with a longer radius dictates the rotating direction of the ellipse.

$$= w_p e^{i\omega t} + w_m e^{-i\omega t} \quad (5)$$

$$\text{or} \quad W_p e^{i(\omega t + \theta_p)} + W_m e^{-i(\omega t - \theta_m)} \quad (6)$$

where we have introduced a complex conjugate operator notation $*$, and

$$w_p \equiv W_p e^{i\theta_p} = \frac{\tilde{u} + i\tilde{v}}{2} \quad (7)$$

$$w_m \equiv W_m e^{i\theta_m} = \left(\frac{\tilde{u} - i\tilde{v}}{2} \right)^* \quad (8)$$

$$\tilde{u} = a_u e^{-i\phi_u} \quad (9)$$

$$\tilde{v} = a_v e^{-i\phi_v} \quad (10)$$

$\tilde{u}$ and $\tilde{v}$ are known as u- and v- complex amplitudes respectively, and the minus signs in the above definitions mean that positive (negative) algebraic values of ϕ_u and ϕ_v themselves represent phase lags (leads) — a tidal convention.

Thus, we have decomposed an ellipse into two circular components: the term with $e^{i\omega t}$ in eq. 4 (or eq. 6) describes an anti-clockwise circle with a radius of W_p , and the term with $e^{-i\omega t}$ describes a clockwise circle with a radius of W_m (figure 1). Depending on whether W_p is greater than, equal to or less than W_m , the ellipse will traverse either anti-clockwise, rectilinear, or clockwise.

When the two circular components are aligned in the same direction, the tidal current will reach its maximum. From, eq. 6, we can see that will happen when

$$\omega t + \theta_p = -\omega t + \theta_m + 2k\pi \quad (11)$$

where integer $k = 0, \pm 1, \pm 2, \pm 3, \dots$. Denote t_{max} as a t satisfying the above criterion, then the phase angle as introduced above is ωt_{max} , which is given by

$$\text{PHA} = \omega t_{max} = \frac{\theta_m - \theta_p}{2} + k\pi \quad (12)$$

We need only to take its minimum value (i.e., when $k = 0$).

Substitute eq. 12 into eq. 6, we can have a current vector whose length is maximum,

$$\begin{aligned} w_{max} &= W_p e^{i(\omega t_{max} + \theta_p)} + W_m e^{-i(\omega t_{max} - \theta_m)} \\ &= W_p e^{i\left(\frac{\theta_m + \theta_p}{2} + k\pi\right)} + W_m e^{i\left(\frac{\theta_m + \theta_p}{2} - k\pi\right)} \\ &= (W_p + W_m) e^{i\frac{\theta_m + \theta_p}{2}} \quad \text{since } e^{ik\pi} = e^{-ik\pi} = 1 \end{aligned} \quad (13)$$

Thus, the maximum current, or semi-major axis (SEMA), is

$$\text{SEMA} = |w_{max}| = W_p + W_m \quad (14)$$

and its direction, or the inclination, is

$$\text{INC} = \arg(w_{max}) = \frac{\theta_m + \theta_p}{2}. \quad (15)$$

When the two circular components are aligned in opposite directions, i.e.,

$$\omega t + \theta_p = -\omega t + \theta_m + (2k + 1)\pi \quad (16)$$

then the tidal current reaches minimum in its speed. At this time, $t = t_{min}$

$$\omega t_{min} = \frac{\theta_m - \theta_p}{2} + \left(k + \frac{1}{2}\right)\pi \quad (17)$$

and

$$\begin{aligned} w_{min} &= W_p e^{i\left(\frac{\theta_m + \theta_p}{2} + \left(k + \frac{1}{2}\right)\pi\right)} + W_m e^{i\left(\frac{\theta_m + \theta_p}{2} - \left(k + \frac{1}{2}\right)\pi\right)} \\ &= W_p e^{i\left(\frac{\theta_m + \theta_p}{2}\right)} e^{i\frac{\pi}{2}} + W_m e^{i\left(\frac{\theta_m + \theta_p}{2}\right)} e^{-i\frac{\pi}{2}} \\ &= (W_p + W_m e^{-\pi}) e^{i\left(\frac{\theta_m + \theta_p}{2} + \frac{\pi}{2}\right)} \\ &= (W_p - W_m) e^{i\left(\frac{\theta_m + \theta_p}{2} + \frac{\pi}{2}\right)} \end{aligned} \quad (18)$$

therefore, the minimum speed of the tidal current, or semi-minor axis (SEMI) is

$$\text{SEMI} = |w_{\min}| = W_p - W_m \quad (19)$$

and its angle is

$$\arg(w_{\min}) = \frac{\theta_m + \theta_p}{2} + \frac{\pi}{2} \quad (20)$$

Thus, the eccentricity, ECC, is

$$\text{ECC} = \frac{\text{SEMI}}{\text{SEMA}} = \frac{W_p - W_m}{W_p + W_m} \quad (21)$$

When $W_m > W_p$, ECC is negative and the ellipse rotates clock-wisely.

The above is the conversion from ap-parameter to e-parameter. Now consider the inverse: given the four e-parameters of SEMA, ECC, INC, and PHA, how can we recover ap-parameters of a_u , ϕ_u , a_v and ϕ_v ?

As a middle step, we need to recover W_p , θ_p , w_p , W_m , θ_m and w_m . From eqs. 14, 19 and 21, we can have

$$W_p = \frac{1 + \text{ECC}}{2} \text{SEMA} \quad (22)$$

$$W_m = \frac{1 - \text{ECC}}{2} \text{SEMA} \quad (23)$$

and from eqs. 12 (when $k = 0$) and 15, we can have

$$\theta_p = \text{INC} - \text{PHA} \quad (24)$$

$$\theta_m = \text{INC} + \text{PHA} \quad (25)$$

Hence we can know

$$w_p = W_p e^{-i\theta_p} \quad (26)$$

$$w_m = W_m e^{i\theta_m} \quad (27)$$

We then can know further from eqs. 7 ~ 10 that

$$a_u e^{-i\phi_u} = w_p + w_m^* \quad (28)$$

$$a_v e^{-i\phi_v} = \frac{1}{i} (w_p - w_m^*) \quad (29)$$

$$\stackrel{\text{or}}{=} -i (w_p - w_m^*) \quad (30)$$

So,

$$\begin{aligned} a_u &= |w_p + w_m| \\ \phi_u &= -\arg(w_p + w_m) \end{aligned} \quad (31)$$

and,

$$a_v = |(w_p - w_m^*)| \quad (32)$$

$$\phi_v = -\arg\left(\frac{w_p - w_m^*}{i}\right) \quad (33)$$

1.2 Decoupling of the linear tidal momentum equations

Consider

$$\frac{\partial u}{\partial t} - fv = -g \frac{\partial \eta}{\partial x} + \frac{\partial}{\partial z} \left(\nu \frac{\partial u}{\partial z} \right) \quad (34)$$

$$\frac{\partial v}{\partial t} + fu = -g \frac{\partial \eta}{\partial y} + \frac{\partial}{\partial z} \left(\nu \frac{\partial v}{\partial z} \right) \quad (35)$$

where all the variables are real and other notations are hopefully standard to you. By adding eq. 34 and $i \times$ eq. 35, and using w defined by eq. 3, we can merge the above two equations into the following complex one,

$$\frac{\partial w}{\partial t} + ifw = -g \nabla \eta + \frac{\partial}{\partial z} \left(\nu \frac{\partial w}{\partial z} \right) \quad (36)$$

where

$$\nabla \equiv \frac{\partial}{\partial x} + i \frac{\partial}{\partial y}. \quad (37)$$

Assume u and v of the forms given in eqs. 1 and 2, and similarly for η , i.e.,

$$\eta = a_\eta \cos(\omega t - \phi_\eta) \quad (38)$$

which can be split into two circular parts as we did for u and v ,

$$\begin{aligned} \eta &= a_\eta \cos(\omega t - \phi_\eta) \\ &= \frac{a_\eta e^{-i\phi_\eta}}{2} e^{i\omega t} + \frac{a_\eta e^{i\phi_\eta}}{2} e^{-i\omega t} \\ &= \eta_p e^{i\omega t} + \eta_m e^{-i\omega t}, \end{aligned} \quad (39)$$

where the tidal convention to define a complex variable is used again, i.e.,

$$\eta_p = \frac{a_\eta e^{-i\theta_\eta}}{2} \quad (40)$$

$$\eta_m = \frac{a_\eta e^{i\theta_\eta}}{2} \quad (41)$$

$$\stackrel{or}{=} \eta_p^*. \quad (42)$$

Using the above equation and eqs. 5 we can rewrite eq. 36 as

$$\left[i(f + \omega)w_p + g \nabla \tilde{\eta}_p - \frac{\partial}{\partial z} \left(\nu \frac{\partial w_p}{\partial z} \right) \right] e^{i\omega t} + \left[i(f - \omega)w_m + g \nabla \tilde{\eta}_m - \frac{\partial}{\partial z} \left(\nu \frac{\partial w_m}{\partial z} \right) \right] e^{-i\omega t} = 0 \quad (43)$$

Since $e^{i\omega t}$ and $e^{-i\omega t}$ are linearly independent of each other, for the above equation to hold, their coefficients must be zero, i.e.,

$$i(f + \omega)w_p = -g \nabla \eta_p + \frac{\partial}{\partial z} \left(\nu \frac{\partial w_p}{\partial z} \right), \quad (44)$$

$$i(f - \omega)w_m = -g \nabla \eta_m + \frac{\partial}{\partial z} \left(\nu \frac{\partial w_m}{\partial z} \right). \quad (45)$$

In the literature, you may find the following form of equation for w_m ,

$$-i(f - \omega)w_m^* = -g \nabla^* \eta_p + \frac{\partial}{\partial z} \left(\nu \frac{\partial w_m^*}{\partial z} \right). \quad (46)$$

(where the conjugate signs on w_m may then be dropped) Eqs. 45 and 46 are equivalent, for they can be obtained from taking conjugate on each other.

1.3 Solutions to w_p and w_m when ν is depth invariant

Subject the decoupled equations of 44 and 45 to the following boundary conditions

$$\nu \frac{\partial(w_p, w_m)}{\partial z} = (0, 0) \quad \text{at } z = 0 \quad (47)$$

$$\nu \frac{\partial(w_p, w_m)}{\partial z} = \kappa(w_p, w_m) \quad \text{at } z = -h(x, y) \quad (48)$$

where κ is a bottom frictional parameter. The adoption of this parameter allows us to accommodate two types of bottom conditions: 1) slip conditions: $(w_p, w_m)_{z=-h} \neq (0, 0)$, this is achieved by setting κ as a finite but non-zero number, in this case, if there is, and shall be, a certain size of bottom stress, there will be non-zero bottom velocities, which means flows are allowed to have motion relative to the “bottom” (in this case, the bottom is understood as the top of the bottom boundary log-layer); 2)

sticky bottom conditions, i.e., $(w_p, w_m)_{z=-h} = (0, 0)$, in this case flow is not allowed to have motion relative to the bottom (the real sea bottom), this condition can be achieved by setting κ to be infinitely large.

The solutions to w_p and w_m can be found as

$$w_p = \text{BE}_p \nabla \eta_p \quad (49)$$

$$w_m = \text{BE}_m \nabla \eta_m \quad (50)$$

where the two-letter variables BE_p and BE_m stand for two **B**ottom **E**kman spirals in rotating components, they spiral near the bottom (just like wind driven surface Ekman spirals near the sea surface) and approach to geostrophic flow in the interior. Their details are expressed in the following:

$$\text{BE}_p(x, y, z) = -\frac{g}{\nu\alpha^2} \left[1 - \frac{\cosh \alpha z}{\cosh \alpha h + \frac{\alpha\nu}{\kappa} \sinh \alpha h} \right] \quad (51)$$

$$\text{BE}_m(x, y, z) = -\frac{g}{\nu\beta^2} \left[1 - \frac{\cosh \beta z}{\cosh \beta h + \frac{\beta\nu}{\kappa} \sinh \beta h} \right] \quad (52)$$

where

$$\alpha = \frac{1+i}{\delta_e} \sqrt{1 + \frac{\sigma}{f}} \quad (53)$$

$$\beta = \frac{1+i}{\delta_e} \sqrt{1 - \frac{\sigma}{f}} \quad (54)$$

$$\delta_e = \sqrt{\frac{2\nu}{f}} \quad (\text{Ekman depth}). \quad (55)$$

The solution to eq. 46 is simply as

$$w_m^* = \text{BE}_m^* \nabla^* \eta_p \quad (56)$$

where BE_m^* is the conjugate of BE_m given by eq. 52. (Also note function $\cosh$ has a property of $(\cosh \alpha z)^* = \cosh(\alpha^* z)$).

As $f \rightarrow \sigma$, $\beta \rightarrow 0$ and BE_m becomes indefinite (0/0 type). This can be the situation for the northern hemisphere, as I have been encountered with recently in my Canadian Arctic archipelago M_2 tidal data assimilation project. Similar concern exist for BE_p for the southern hemisphere where $f + \sigma$ (hence α) can be zero. This type singularity may be referred as inertial frequency singularity. The singularity is not essential but we have to remove it before we can feed the formula into the computer,

for otherwise the computer will generate garbage since there are only limited significant numbers used in a computer. This can be done by expanding eqs. 53 and 54 in power series:

$$\text{BE}_p(x, y, z) = C \sum_{t=1}^{\infty} \left[\frac{1 - (z/h)^{2t}}{2t} + \frac{\nu}{\kappa h} \right] \frac{(\alpha h)^{2t-2}}{(2t-1)!} \quad (57)$$

$$\text{BE}_m(x, y, z) = D \sum_{t=1}^{\infty} \left[\frac{1 - (z/h)^{2t}}{2t} + \frac{\nu}{\kappa h} \right] \frac{(\beta h)^{2t-2}}{(2t-1)!} \quad (58)$$

where

$$C = -\frac{gh^2}{\nu \left(1 + \frac{\alpha\nu}{\kappa} \tanh \alpha h\right)} \frac{2e^{-\alpha h}}{1 + e^{-2\alpha h}} \quad (59)$$

$$D = -\frac{gh^2}{\nu \left(1 + \frac{\beta\nu}{\kappa} \tanh \beta h\right)} \frac{2e^{-\beta h}}{1 + e^{-2\beta h}} \quad (60)$$

Assume the ocean depth, h , is always finite and let us now consider a case where $\alpha h \rightarrow 0$, which will arise when either or both i) $f \rightarrow -\sigma$ or ii) $\nu \rightarrow \infty$ happens. In this limiting case,

$$\lim_{\alpha \rightarrow 0} \text{BE}_p(x, y, z) = \begin{cases} -\left[\frac{g}{\nu} \frac{h^2 - z^2}{2} + \frac{gh}{\kappa} \right] & \text{when } \nu \text{ is finite} \\ -\frac{gh}{\kappa} & \text{when } \nu \text{ is infinite} \end{cases} \quad (61)$$

Likewise,

$$\lim_{\beta \rightarrow 0} \text{BE}_m(x, y, z) = \begin{cases} -\left[\frac{g}{\nu} \frac{h^2 - z^2}{2} + \frac{gh}{\kappa} \right] & \text{when } \nu \text{ is finite} \\ -\frac{gh}{\kappa} & \text{when } \nu \text{ is infinite} \end{cases} \quad (62)$$

Thus we have resolved the apparent inertial frequency singularity.

If use the ratio test, we will see that the radii of convergence of the two series are infinite. Hence for any significantly large values of αh and βh , the series of eqs. 57 and 58 will converge to the same values of its finite forms of eqs. 51 and 52. However we might use the finite forms themselves for the sake of convergence rate. A MATLAB program, called cBEpm, is listed in the following section for calculating BE_p and BE_m . You will see in that program that when $|\alpha h|$ is less than 1, the series form is used, otherwise, the finite form is used. (Even when αh goes up to 10, experiments show the series converges fast enough.) Why is it named as cBEpm? This is because I have lived with the combination of eqs. 44 and 45 and have made a program called BEpm and used it everywhere for calculating BE_p and BE_m^* . So to me “c” in cBEpm means conjugate, but to you it may be interpreted as a reminder that BEp and BEm are complex valued. (You do not want to punish me by modifying many of my other programs, do you?).

2 Programs

Three programs are included here: ap2ep.m, which converts ap-parameters to e-parameters, ep2ap.m which is the inverse of ap2ep.m, and cBEpm.m for calculating the bottom Ekman spirals (BE_p and BE_m). See the comments in the programs for more details.

2.1 ap2ep.m

```
function [SEMA, ECC, INC, PHA, w]=ap2ep(Au, PHIu, Av, PHIV, plot_demo)
%
% Convert tidal amplitude and phase lag (ap-) parameters into tidal ellipse
% (e-) parameters. Please refer to ep2ap for its inverse function.
%
% Usage:
%
% [SEMA, ECC, INC, PHA, w]=ap2ep(Au, PHIu, Av, PHIV, plot_demo)
%
% where:
%
% Au, PHIu, Av, PHIV are the amplitudes and phase lags (in degrees) of
% u- and v- tidal current components. They can be vectors or
% matrices or multidimensional arrays.
%
% plot_demo is an optional argument, when it is supplied as an array
% of indices, say [i j k l], the program will plot an ellipse
% corresponding to Au(i,j, k, l), PHIu(i,j,k,l), Av(i,j,k,l), and
% PHIV(i,j,k,l);
%
% Any number of dimensions are allowed as long as your computer
% resource can handle.
%
% SEMA: Semi-major axes, or the maximum speed;
% ECC: Eccentricity, the ratio of semi-minor axis over
% the semi-major axis; its negative value indicates that the ellipse
% is traversed in clockwise direction.
% INC: Inclination, the angles (in degrees) between the semi-major
% axes and u-axis.
% PHA: Phase angles, the time (in angles and in degrees) when the
% tidal currents reach their maximum speeds, (i.e.
% PHA=omega*tmax).
%
% These four e-parameters will have the same dimensionality
```

```

%           (i.e., vectors, or matrices) as the input ap-parameters.
%
%   w:   Optional. If it is requested, it will be output as matrices
%        whose rows allow for plotting ellipses and whose columns are
%        for different ellipses corresponding columnwise to SEMA. For
%        example, plot(real(w(1,:)), imag(w(1,:))) will let you see
%        the first ellipse. You may need to use squeeze function when
%        w is a more than two dimensional array. See example.m.
%
% Document:   tidal_ellipse.ps

if nargin < 5
    plot_demo=0; % by default, no plot for the ellipse
end

% Assume the input phase lags are in degrees and convert them in radians.
PHIu = PHIu/180*pi;
PHIv = PHIv/180*pi;

% Make complex amplitudes for u and v
i = sqrt(-1);
u = Au.*exp(-i*PHIu);
v = Av.*exp(-i*PHIv);

% Calculate complex radius of anticlockwise and clockwise circles:
wp = (u+i*v)/2;      % for anticlockwise circles
wm = conj(u-i*v)/2;  % for clockwise circles
% and their amplitudes and angles
Wp = abs(wp);
Wm = abs(wm);
THETAp = angle(wp);
THETAm = angle(wm);

% calculate e-parameters (ellipse parameters)
SEMA = Wp+Wm;          % Semi Major Axis, or maximum speed
SEMI = Wp-Wm;          % Semi Minor Axis, or minimum speed
ECC = SEMI./SEMA;      % Eccentricity
PHA = (THETAm-THETAp)/2; % Phase angle, the time (in angle) when
                        % the velocity reaches the maximum
INC = (THETAm+THETAp)/2; % Inclination, the angle between the
                        % semi major axis and x-axis (or u-axis).

% convert to degrees for output
PHA = PHA/pi*180;

```

```

    INC = INC/pi*180;
    THETAp = THETAp/pi*180;
    THETAm = THETAm/pi*180;

% flip THETAp and THETAm, PHA, and INC in the range of
% [-pi, 0) to [pi, 2*pi), which at least is my convention.
    id = THETAp < 0;    THETAp(id) = THETAp(id)+360;
    id = THETAm < 0;    THETAm(id) = THETAm(id)+360;
    id = PHA < 0;       PHA(id) = PHA(id)+360;
    id = INC < 0;       INC(id) = INC(id)+360;

    if nargout == 5
        ndot=36;
        dot=2*pi/ndot;
        ot=[0:dot:2*pi-dot];
        w=wp(:)*exp(i*ot)+wm(:)*exp(-i*ot);
        w=reshape(w, [size(Au) ndot]);
    end

    if any(plot_demo)
        plot_ell(SEMA, ECC, INC, PHA, plot_demo)
    end

```

2.2 ep2ap.m

```

function [Au, PHl, Av, PHv, w]=ep2ap(SEMA, ECC, INC, PHA, plot_demo)
%
% Convert tidal ellipse parameters into amplitude and phase lag parameters.
% Its inverse is ap2ep.m. Please refer to ap2ep for the meaning of the
% inputs and outputs.
%
% Zhigang Xu
% Oct. 20, 2000
%
% Document: tidal_ellipse.ps
%
if nargin < 5
    plot_demo=0; % by default, no plot for the ellipse
end

Wp = (1+ECC)/2 .*SEMA;

```

```

wm = (1-ECC)/2 .*SEMA;
THETAp = INC-PHA;
THETAm = INC+PHA;

%convert degrees into radians
THETAp = THETAp/180*pi;
THETAm = THETAm/180*pi;

%Calculate wp and wm.
wp = Wp.*exp(i*THETAp);
wm = Wm.*exp(i*THETAm);

if nargin == 5
    ndot=36;
    dot = 2*pi/ndot;
    ot = [0:dot:2*pi-dot];
    w = wp(:)*exp(i*ot)+wm(:)*exp(-i*ot);
    w=reshape(w, [size(wp) ndot]);
end

% Calculate cAu, cAv --- complex amplitude of u and v
cAu = wp+conj(wm);
cAv = -i*(wp-conj(wm));
Au = abs(cAu);
Av = abs(cAv);
PHIu = -angle(cAu)*180/pi;
PHIv = -angle(cAv)*180/pi;

% flip angles in the range of [-180 0) to the range of [180 360).
id = PHIu < 0; PHIu(id) = PHIu(id) + 360;
id = PHIv < 0; PHIv(id) = PHIv(id) + 360;

if any(plot_demo)
    plot_ell(SEMA,ECC,INC,PHA,plot_demo)
end

```

2.3 plot_ell.m

```

function w=plot_ell(SEMA, ECC, INC, PHA, IND)
%
% An auxiliary function used in ap2ep and ep2ap for plotting tidal ellipse.
% The inputs, MA, ECC, INC and PHA are the output of ap2ep and IND is a

```

```

% vector for indices for plotting a particular ellipse, e.g., if IND=[2 3 1];
% the ellipse corresponding to the indices of (2,3,1) will be plotted.
%-----

Size_SEMA=size(SEMA);
len_IND=length(IND);
if IND
    cmd=['sub2ind(size_SEMA' ];
    if len_IND==1
        titletxt=['Ellipse '];
    else
        titletxt=['Ellipse ('];
    end

    for k=1:len_IND;
        cmd=[cmd ',num2str(IND(k))'];
        if k<len_IND
            titletxt=[titletxt num2str(IND(k)) ','];
        elseif len_IND==1
            titletxt=[titletxt num2str(IND(k))'];
        else
            titletxt=[titletxt num2str(IND(k)) ' '];
        end
    end

    cmd=['n=' cmd ');'];
    eval(cmd);

    figure(gcf)
    clf
    do_the_plot(SEMA(n), ECC(n), INC(n), PHA(n));
    title(titletxt);
elseif len_IND
    msg1=['IND input contains zero element(s)!'];
    msg2=['No ellipse will be plotted.'];
    disp(msg1);
    disp(msg2);
end

%-----
%begin of plot subfunction
function w=do_the_plot(SEMA, ECC, INC, PHA)

```

```

SEMI = SEMA.*ECC;
Wp = (1+ECC)/2 .*SEMA;
Wm = (1-ECC)/2 .*SEMA;
THETAp = INC-PHA;
THETAm = INC+PHA;

%convert degrees into radians
THETAp = THETAp/180*pi;
THETAm = THETAm/180*pi;
INC = INC/180*pi;
PHA = PHA/180*pi;

%Calculate wp and wm.
wp = Wp.*exp(i*THETAp);
wm = Wm.*exp(i*THETAm);

dot = pi/18;
ot = [0:dot:2*pi-dot];
a = wp*exp(i*ot);
b = wm*exp(-i*ot);
w = a+b;

wmax = SEMA*exp(i*INC);
wmin = SEMI*exp(i*(INC+pi/2));

plot(real(w), imag(w))
axis('equal');
hold on
plot([0 real(wmax)], [0 imag(wmax)], 'm');
plot([0 real(wmin)], [0 imag(wmin)], 'm');
xlabel('u');
ylabel('v');
plot(real(a), imag(a), 'r');
plot(real(b), imag(b), 'g');
hnd_a=line([0 real(a(1))], [0 imag(a(1))], 'color', 'r', 'marker','o');
hnd_b=line([0 real(b(1))], [0 imag(b(1))], 'color', 'g', 'marker','o');
hnd_w=line([0 real(w(1))], [0 imag(w(1))], 'color', 'b', 'marker','o');
plot(real(a(1)), imag(a(1)), 'ro');
plot(real(b(1)), imag(b(1)), 'go');
plot(real(w(1)), imag(w(1)), 'bo');
hnd_ab=line(real([a(1) a(1)+b(1)]), imag([a(1) a(1)+b(1)]), ...
    'linestyle', '--', 'color', 'g');
hnd_ba=line(real([b(1) a(1)+b(1)]), imag([b(1) a(1)+b(1)]), ...

```

```

        'linestyle', '--', 'color', 'r');

for n=1:length(ot);
    set(hnd_a, 'xdata', [0 real(a(n))], 'ydata', [0 imag(a(n))]);
    set(hnd_b, 'xdata', [0 real(b(n))], 'ydata', [0 imag(b(n))]);
    set(hnd_w, 'xdata', [0 real(w(n))], 'ydata', [0 imag(w(n))]);
    hold on
    plot(real(a(n)), imag(a(n)), 'ro');
    plot(real(b(n)), imag(b(n)), 'go');
    plot(real(w(n)), imag(w(n)), 'bo');
    set(hnd_ab, 'xdata', real([a(n) a(n)+b(n)]), 'ydata', ...
        imag([a(n) a(n)+b(n)]))
    set(hnd_ba, 'xdata', real([b(n) a(n)+b(n)]), 'ydata', ...
        imag([b(n) a(n)+b(n)]))
end

hold off

%end of plot subfunction
%-----
%
%
```

2.4 example.m

```

%demonstrate how to use ap2ep and ep2ap

Au=rand(4,3,2);          % so 4x3x2 multi-dimensional matrices are used for the
Av=rand(4,3,2);          % demonstration.
Phi_v=rand(4,3,2)*360;   % phase lags inputs are expected to be in degrees.
Phi_u=rand(4,3,2)*360;

figure(1)
clf
[SEMA ECC INC PHA w]=ap2ep(Au, Phi_u, Av, Phi_v, [2 3 1]);

figure(2)
clf
[rAu rPhi_u rAv rPhi_v rw]=ep2ap(SEMA, ECC, INC, PHA, [2 3 1]);

%check if ep2ap has recovered Au, Phi_u, Av, Phi_v
max(max(max(abs(rAu-Au))))          % = 9.9920e-16
max(max(max(abs(rAv-Av))))          % = 6.6613e-16
```

```

max(max(max(abs(rPhi_u-Phi_u))))      % = 4.4764e-13
max(max(max(abs(rPhi_v-Phi_v))))      % = 1.1369e-13
max(max(max(max(abs(w-rw))))))        % = 1.3710e-15
% for the random realization I had, the differences are listed on the right
% hand of the above column. What are yours?

% The above example function calls have already plotted an ellipse for you.
% To plot an ellipse separately, you may do
%
figure(3)
clf
plot(real(squeeze(w(2,3,1,:))), imag(squeeze(w(2,3,1,:))));

%here squeeze is needed because w is a multiple dimensional array.

```

2.5 cBEpm.m

```

function [BEp, BEm]=cBEpm(g, f, sigma, nu, kappa, z, h)
%
%Evaluate the theoretical vertical profiles (or Bottom Ekman spiral profiles)
%of tidal currents in the two rotary directions driven by half-unit of sea
%surface gradients in the two directions respectively. Eddy viscosity is
%assumed as vertically invariant. See tidal_ellipse.ps for more details.
%
%
%inputs:
%
%      g, the gravity acceleration,
%      f, the Coriolis parameter,
%      nu, the eddy viscosity
%      kappa, the bottom frictional coefficient
%      z, the vertical coordinates, can be a vector but must be
%          within [0 -h];
%      h, the water depth, must be positive.
%
%      Note: except for z, all other inputs must be scalars.
%
%outputs:
%
%      BEp and BEm, the same dimensions of z, the outputs for the vertical
%          velocity profiles driven respectively by a unit of sea

```

```

%           surface slope in the positive rotation direction and negative
%           rotation direction for when the eddy viscosity is vertically
%           invariant. See the associated document for more details.

if length(g)>1 | length(f)>1 | length(sigma)>1 | ...
    length(nu)>1 | length(kappa)>1 | length(h)>1
    error('inputs of g,f,sigma, nu, kappa, and h should be all scalars!')
end

if any(z/h>0) | any(z/h<-1)
    disp('z must be negative and must be within [0 -h]')
end

delta_e = sqrt(2*nu/f); %Ekman depth
alpha = (1+i)/delta_e*sqrt(1+sigma/f);
beta = (1+i)/delta_e*sqrt(1-sigma/f);

BEp = get_BE(g, alpha, h, z, nu, kappa);
BEm = get_BE(g, beta, h, z, nu, kappa);

%subfunction
%-----
function BE=get_BE(g, alpha, h, z, nu, kappa)

    z = z(:);
    z_h = z/h;
    ah = alpha*h;
    az = alpha*z;
    ah2 = ah*2;
    anu_k = alpha*nu/kappa;
    nu_kh = nu/(kappa*h);

    if abs(ah) < 1 %series solution
T = 10;
C = -g*h*h/(nu*(1+anu_k*tanh_v5_2(ah)))*2;
A1 = (1-z_h.*z_h)/2+nu_kh;
B1 = exp(-ah)/(1+exp(-ah2));
B = B1;
    series_sum=A1*B1;

for t = 2:T;
    t2=2*t;

```

```

        A = (1-z_h.^t2)./t2+nu_kh;
        B = B*ah*ah/(t2-1)/(t2-2);
        series_sum = series_sum+A*B;
    end

    BE = C*series_sum;

    else      %finite solution

        c = -g*h*h/nu;
        denom=(exp(az-ah)+exp(-(az+ah)))/(1+exp(-2*ah));
                % =cosh(az)/cosh(ah);
                %but this a better way to evaluate it.
        numer=1+anu_k*tanh_v5_2(ah);
        BE = c*((1-denom/numer)/(ah*ah));
    end

%end of subfunction
%
%Note tanh_v5_2 is a copy of tanh from Matlab v5.2, which has worked well!
%It seems that Matlab v5.3 has some bug(s) in tanh function! It cannot deal
%with large argument. try z=7.7249e02*(1+i), tanh(z) and tanh_v5_2(z) to
%see the difference.

```